order29 RESULTS

PREREG sha256 abf67d465337c6a44fa933341b51418d279a96e84f54a759a183a932fb3baaac (hashed before numerics).

Recovered order17 law

The fixed-M law with a free scalar κ and weight w is r(w) = min_κ ‖S − κ x^{−w}35^{−1/2}MY‖/‖S‖, applied to lines plus completion. For A2 the completion is A2(S^o) = 0, so the object tested is LN.

Controls

C1 corner(K) = CO: 0.0, PASS. C2 ‖R(CO)‖ = 54.406: rel err 0.0, PASS. Iterated subdivision Σ Cor = CO + ¼(2ⁿ−1)vtx: 2.2e−14, 4.3e−14, 9.6e−14 for n = 1, 2, 3, PASS.

RG21 (primary): FAIL

object	non-vac d	r(w=−1)	r(−½)	r(0)	r(½)	r(1)	r_min	verdict
LN	0.998	0.992	0.992	0.993	0.997	0.999	0.992	**FAIL**
Π+LN	0.994	0.9999	0.9999	0.9999	0.9999	1.000	0.9999	FAIL
Π−LN	1.000	0.983	0.982	0.983	0.992	0.998	0.982	FAIL
CO (comparison)	—	0.0585	0.210	0.503	0.848	0.982	0.0585	FAIL

Absolute residuals for LN are ≈105.5 at every weight, with ‖S_LN‖ ≈ 106. Even with κ free, M·LN is essentially uncorrelated with LN(−1/x).

Near-law isolated as corner-only: NO by the preregistered criterion, which needed LN to PASS. Descriptively (Category B), the near-law (r ≈ 0.06 at w = −1, κ ≈ 1) is seen only in the corner. The lines show no law at all (r ≈ 0.99) in either P-sector, so the lines are not a hidden law-obeying part. Prediction FAIL was correct (blind with respect to this metric).

RG22 (secondary): FAIL at all depths (rule fixed in PREREG: Stern–Brocot alternating signs, coefficient 1)

depth n	cones	non-vac d	‖R(Alt_n)‖	‖R(CO+Alt_n)‖	rel to ‖R(CO)‖	|cos(R(Alt_n), −R(CO))|	verdict
1	2	0.580	224.9	219.0	4.02	0.229	FAIL / FAIL
2	4	0.582	418.9	415.7	7.64	0.125	FAIL / FAIL
3	8	0.653	727.2	726.1	13.35	0.060	FAIL / FAIL

‖R(Alt_n)‖ grows with depth and its alignment with −R(CO) decays. The rule is divergent and does not converge to a cancellation. Prediction FAIL was correct (blind).

Category

A: none new (the iterated subdivision identity is v8 Prop. subdiv applied n times). B: all numbers above. v11: no. Ξ: not identified. E1–E3 OPEN.

order29 NOTES

- The lines obey no fixed-M law (κ free, any w ∈ [−1, 1]); the corner is the only near-law object (r ≈ 0.0585). The near-law is a property of A2_corner. This does not satisfy the preregistered "isolation" definition, because that required a PASS for the lines.
- Alternating Stern–Brocot sums diverge in the law metric; cone corners at depth n are not small corrections.
- Lesson: corner terms of finer cones are O(1) each; cancellation rules need weights that decay with depth (e.g. 2^{−n} or arc-length weights).
- Open next step (order30, not started): fix in advance a decaying geometric weight (hyperbolic arc length of each Stern–Brocot cone on the orthant arc, total L = ½log(6+√35), from v8) and test Σ™_i/L·Cor_{C_i} → limit; or test the near-law of CO against the Siegel-arc average of single-ray corner terms.
- Geometric only; MAPS_CANDIDATES/DISCOVER_GRADING_PATH/MARKOV_FIBRE_FREQ untouched; no Resonant objects; no bridge claim; CORE_FREEZE respected. nice, OMP_NUM_THREADS=1, 3 GB, ~165 s; no push or sends; new files only.